



Introduction

- Audio-Visual Speech Recognition** is the task of automatically transcribing speech using acoustic signals and visual cues mirroring **multimodal nature of human perception [1]**
- Current AVSR architectures are mainly Transformer-based and the recent advent of Large Language Models (LLMs) has further advanced performance

However:

- It remains unclear if improvements are mainly due to Audio-based Speech Recognition and LLM innovations
- Whether learned representations align with established linguistic concepts, such as phonemes & visemes

Objective

- Can an AVSR model be explicitly guided to encode viseme information during training, to make it more robust under noisy conditions?

Resources

- Phonemes are the basic speech units and visemes are their visual equivalents
- We employ Lee's phoneme-to-viseme mapping [3]
- We use LRS3 and obtain phoneme alignments using Montreal Forced Aligner
- Each timestamp is converted to a frame and assigned a viseme label
- CTC loss is used for the viseme classification, while Cross-Entropy loss is used for the main speech recognition task
- The secondary task is introduced from the start of fine-tuning

Architecture

- Adapted AV-HuBERT [2] to include a small viseme classification head, adding only 1.32% extra parameters
- The introduction of the secondary task guides the encoder to preserve visual articulatory information
- The proposed architecture has been tested on Base and Large AV-HuBERT variants.

Results

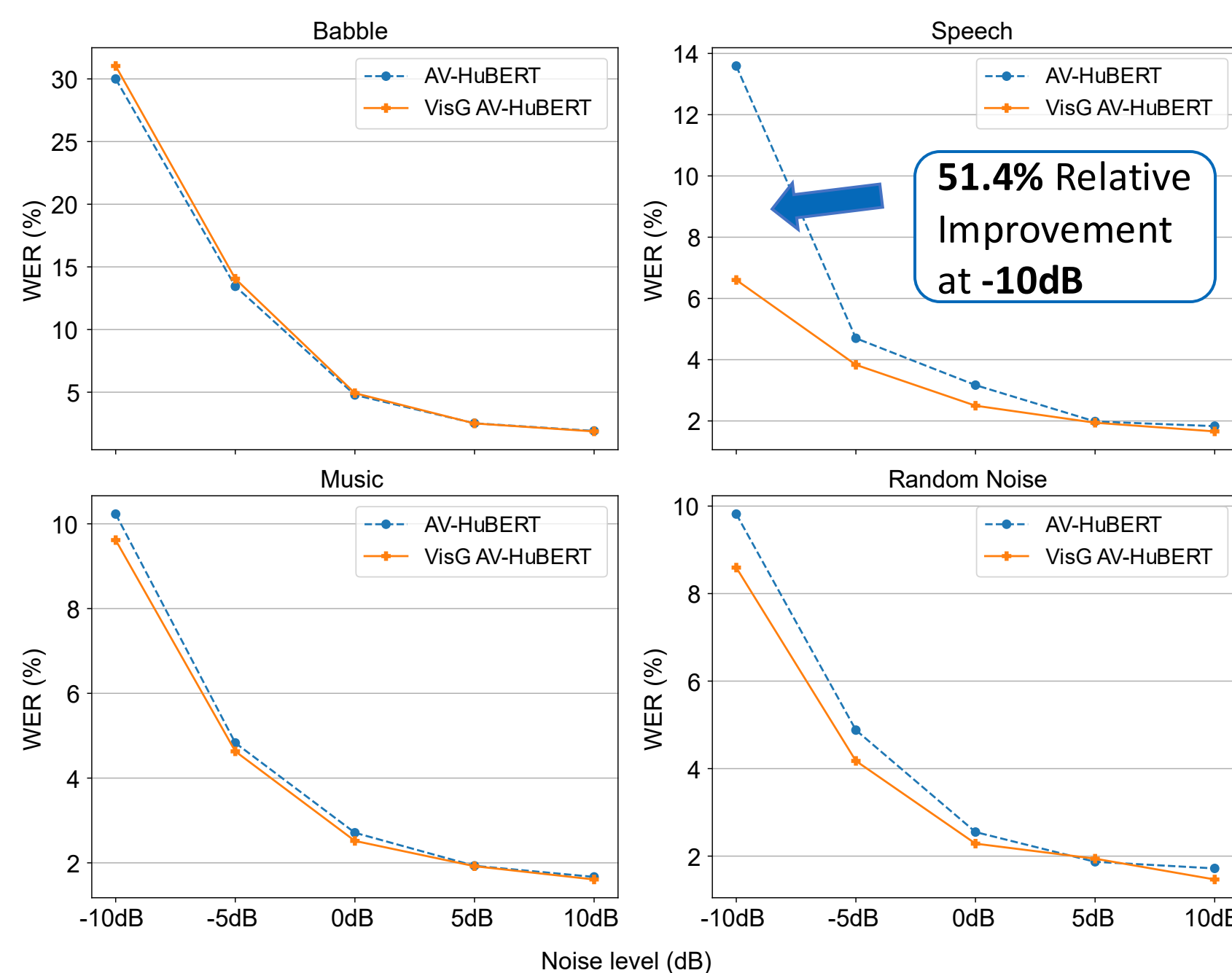
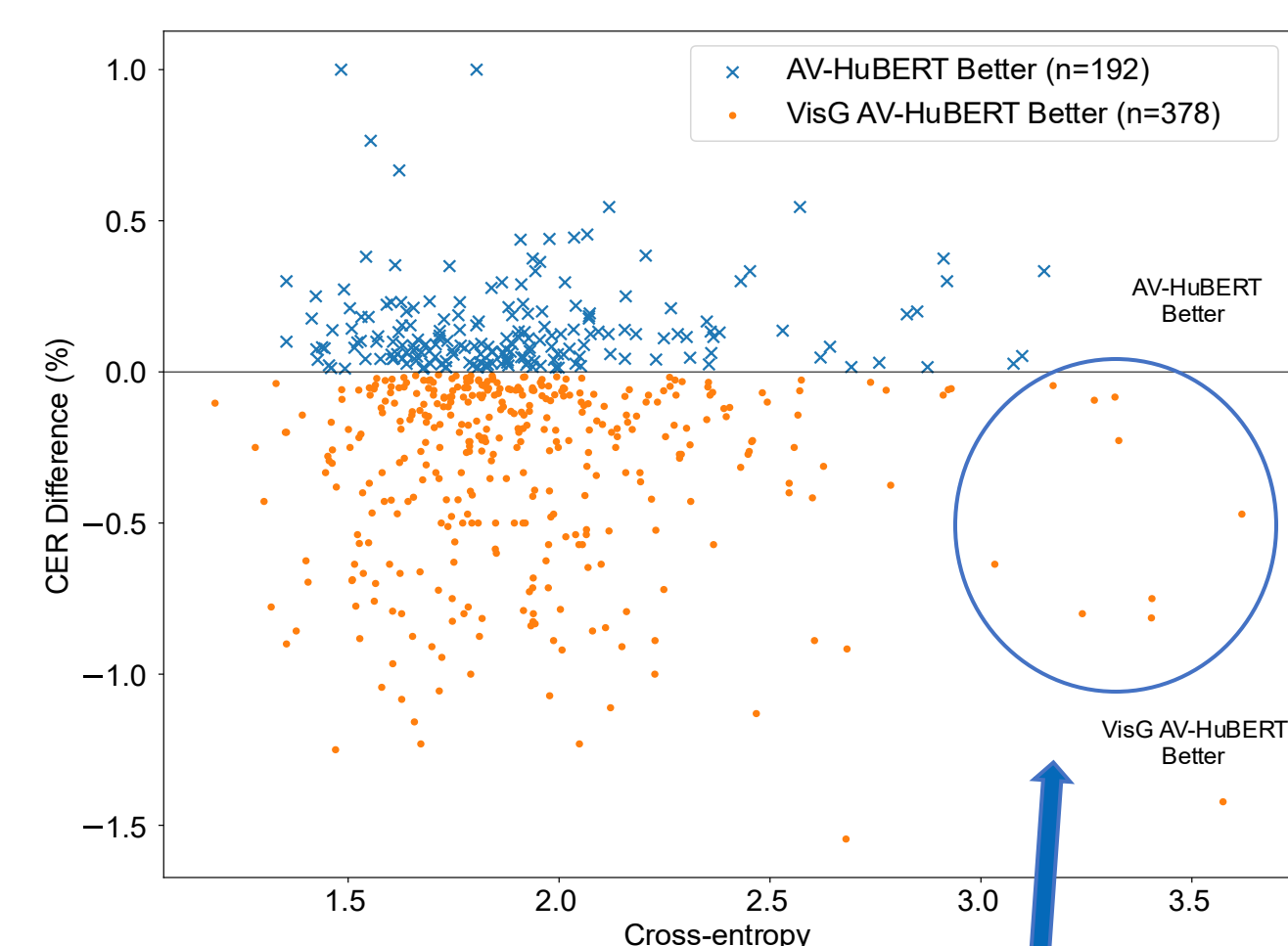


Table 3.1: WER (%) Comparison under Clean Conditions

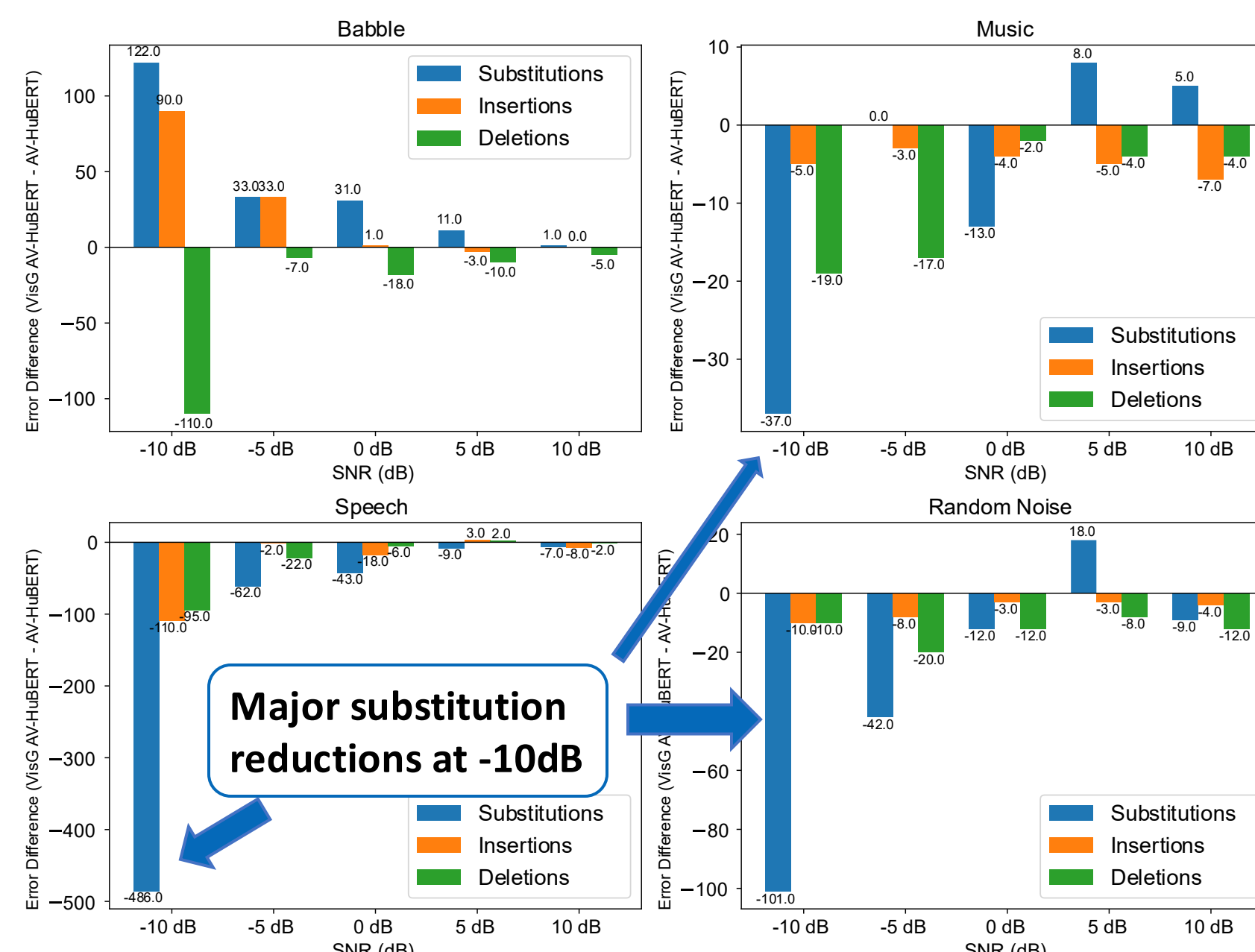
Model	Params	WER LRS3 ↓	WER LRS2 ↓
AV-HuBERT Base	160M	4.1	10.09
VisG AV-HuBERT Base	162M	3.78*	10.58
AV-HuBERT Large	477M	1.4	10.3
VisG AV-HuBERT Large	484M	1.38	9.925

Minor improvements in clean conditions, in line with our expectations



Our proposed method improves on CER performance on more complex sentences

Viseme	Phoneme	Viseme	Phoneme
F	/f/ /v/	AA	/aa/ /aw/ /ay/
W	/r/ /w/	AH	/ah/
P	/b/ /p/ /m/	AO	/ao/ /oy/ /ow/
K	/g/ /k/ /ng/ /n/ /l/ /y/ /hh/	UH	/uh/ /uw/
T	/t/ /d/ /s/ /z/ /dh/ /th/	ER	/er/
CH	/ch/ /jh/ /sh/ /zh/	EH	/eh/ /ey/ /ae/
/sil/	/sil/	IY	/iy/ /ih/



Significant error reduction in absolute counts, for most cases, especially substitution errors, denoting **improved unit level discrimination**

Discussion & Conclusion

- We have proposed a multi-task training framework to improve performance under noisy conditions
- Our results show that our method is most effective under speech noise and at severe noise levels

References

- [1] W. H. Sumby and Irwin Pollack. Visual contribution to speech intelligibility in noise. The Journal of the Acoustical Society of America, 26(2):212–215, 03 1954.
- [2] B. Shi, W.-N. Hsu, K. Lakhotia, and A. Mohamed, “Learning audio-visual speech representation by masked multimodal cluster prediction,” in International Conference on Learning Representations, 2022.
- [3] S. Lee and D. Yook, “Audio-to-visual conversion using hidden markov models,” in PRICAI 2002: Trends in Artificial Intelligence, Springer, 2002, pp. 563–570.

Acknowledgment

This publication has emanated from research conducted with the financial support of Taighde Éireann – Research Ireland under Grant number 22/FFP-A/11059.